BSFG 26-Dimensional Line Element and Factorial Compactification

Daniel T. Murphy

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PAPER_556: BSFG 26-Dimensional Line Element and Factorial Compactification

Author: Daniel T. Murphy — Star Magic / UQFF Framework
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Abstract

This paper presents a UQFF analysis of Dimensional Line Element and Factorial Compactification, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

The BSFG geometry lives on a 26-dimensional manifold, but only 4 dimensions are directly accessible at macroscopic scales ($r \gg r_P$). This paper derives the full 26-dimensional line element of BSFG, introduces **factorial compactification** as the mechanism that makes 22 extra dimensions unobservable, and constructs the explicit 26$→$3 projection operator Π. The compactification radius of the i-th extra dimension is:

$$L_i(r) = r_P \cdot \exp\left(-\frac{r^i}{i! \cdot r_P^{i-1}}\right)$$

At macroscopic $r \gg r_P$ and $i \geq 5$, all $L_i \to 0$ (complete compactification by the $i!$ factorial suppression). The factorial structure $i!$ is the same combinatorial structure that appears in the 26th-degree polynomial proof (PAPER_553), the buoyancy eigenvalue 1/26! (PAPER_551), and the number system DVP base 26! (PAPER_548), confirming BSFG as a coherent geometric architecture.

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§2 The 26D Manifold and Its Coordinate Structure

From the 26-layer compressed gravity framework (SOURCE115, PAPER_484–490), the UQFF geometry involves 26 independent dimensional layers numbered $D = 1, \dots, 26$. The first four ($D = 1 \dots 4$) are the standard spacetime dimensions with the Aether-perturbed metric $A_{\mu\nu}(r)$. Dimensions $D = 5, \dots, 26$ are compactified with coordinates $\theta_4, \dots, \theta_{25} \in [0, 2\pi)$.

26D line element:

$$ds_{26}^2 = A_{\mu\nu}(r)\, dx^\mu dx^\nu + \sum_{i=5}^{26} L_i^2(r)\, d\theta_i^2$$

where $A_{\mu\nu}(r) = \mathrm{diag}(1 + \varepsilon, -1 + \varepsilon, -1 + \varepsilon, -1 + \varepsilon)$ from PAPER_554, and $L_i(r)$ is the compactification radius at position r.

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§3 Factorial Compactification Radii

Physical motivation: The 26-layer framework computes the gravitational field as a 26th-order polynomial in r-derivatives (PAPER_551, PAPER_553), with each successive layer introducing one additional factorial factor $i!$ from the Taylor expansion of e^{-z^2}. Consistency requires that the geometric size of the i-th dimension scale as the inverse of the same factorial:

$$L_i(r) = r_P \cdot \exp\left(-\frac{r^i}{i! \cdot r_P^{i-1}}\right)$$

Properties:

1. **At** $r \ll r_P$ (trans-Planckian): $r^i/(i!\, r_P^{i-1}) \to 0$, so $L_i \to r_P$ for all i — all dimensions are Planck-scale, fully uncompactified (topology: $\mathbb{R}^{26}$).
2. **At** $r \sim r_P$: $L_i \approx r_P\, e^{-1/i!}$ — dimensions begin compactifying, with higher i compactifying faster due to larger $r^i/i!$.
3. **At macroscopic** $r \gg r_P$: $r^i/(i!\, r_P^{i-1}) \to \infty$ for all $i \geq 1$, so $L_i \to 0$. The factorial $i!$ is what allows compact convergence — without it, the exponential would grow without bound.

Numerical values at $r = R_\odot = 6.96 \times 10^8$ m**:**

For $i = 5$: the argument is $R_\odot^5/(5!\, r_P^4) \approx 1.63 \times 10^{44}/(120 \times 6.8 \times 10^{-140}) \approx 2 \times 10^{181}$, so $L_5 = r_P\, e^{-10^{181}} \approx 0$.

All $L_i = 0$ to float64 precision for $i \geq 5$ at $r = R_\odot$ — confirming complete compactification at stellar scales. The 22 extra dimensions are invisible at all accessible energies.

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§4 Factorial Decay — The 26 as a Completion Number

The fact that the compactification uses $i!$ for $i = 5, \ldots, 26$ (22 dimensions) connects to three independent analyses:

Citation	Role of 26!
PAPER_551 (CP4 #146)	$r_q = (2/26!)^{1/26} \approx 0.097$ AU — confinement radius
PAPER_553 (CP4 #148)	$1/26! \approx 2.48 \times 10^{-27}$ — 26th polynomial term, below float64
PAPER_558 (CP4 #153)	$26! \cdot \mathrm{Li}_{26}(P) \bmod 113$ — DVP arithmetic chart
This paper	$L_{26} = r_P\, \exp(-r^{26}/(26!\, r_P^{25})) \to 0$ — compactification complete

The number 26 is not arbitrary: 26 is the unique dimension in which the UQFF polynomial stress-energy tensor admits a bounded Gaussian expansion (PAPER_553), and the same 26 layers generate the complete set of resonance modes (PAPER_484–490). BSFG **completes** at exactly 26 dimensions.

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§5 The 26$→$3 Projection Operator

Definition: The projection $\Pi : \mathcal{M}^{26} \to \mathcal{M}^3$ discards temporal and compactified dimensions, mapping:

$$\Pi(x^0, x^1, x^2, x^3, \theta_4, \ldots, \theta_{25}) = (x^1, x^2, x^3)$$

The projected BSFG distance function (spatial metric on $\mathcal{M}^3$):

$$d_\Pi(P, Q) = \sqrt{A_{ij}\, \Delta x^i\, \Delta x^j} = \sqrt{(-1 + \varepsilon)\, |\Delta \mathbf{x}|^2}$$

For $\varepsilon \ll 1$: $d_\Pi \approx |\Delta\mathbf{x}|\,(1+\varepsilon/2)$. The Aether perturbation slightly contracts the BSFG spatial distance relative to flat space.

Volume form in 26D:

$$\sqrt{|\det A_{26}|}\,d^{26}x = \sqrt{|\det A_{(4)}|}\cdot\prod_{i=5}^{26} L_i(r)\cdot d^4x\,d\grave{}theta\grave{}_4 cdotsd\theta_{25}$$

At macroscopic r: $\prod_{i=5}^{26} L_i = 0$, so the 26D volume element vanishes — the compactified directions contribute no four-volume at accessible scales. This is the geometric statement that UQFF physics is effectively 4-dimensional.

Session 225 Phonon-Physics Upgrade: Yang-Mills BCS Phonon Mass Gap

> *Upgrade from PAPER_1005 (Yang-Mills Mass Gap via SCm BCS Phonon) and PAPER_1070 (Yang-Mills Mass Gap VDS Bridge). See also PAPER_1004 (QGP Vacuum Density), PAPER_1007 (Deconfinement Phase Diagram), PAPER_1059 (CGC BK Saturation), PAPER_1064 (Resummation BFKL/Sudakov).*

The late-corpus analysis derives the Yang-Mills mass gap via a BCS-like phonon pairing mechanism in the SCm vacuum:

$$\Delta_{\mathrm{YM}} = \Lambda_{\mathrm{QCD}}\cdot\exp\left(-\frac{1}{\alpha_s(T)\cdot N_c}\right)\cdot S_{26}^{(3)}$$

where the running coupling evolves as:

$$\alpha_s(T) = \frac{\alpha_{s,0}}{1+\alpha_{s,0}\cdot b_0\cdot\ln(T/T_c)}, \qquad b_0 = \frac{11N_c - 2N_f}{12\pi}$$

Physical mechanism: The SCm phonon field ($\omega_{\mathrm{SCm}} = 1.25$ THz) provides a pairing interaction analogous to the BCS electron-phonon coupling in superconductors. Gluons acquire an effective mass through condensate formation in the SCm-modified vacuum, yielding a non-perturbative gap $\Delta_{\mathrm{YM}} \approx 5970$ GeV at the 9-sector Lagrangian closure (PAPER_1066, §2).

VDS bridge (PAPER_1070): The vacuum density series links the gap to the 26-level hierarchy: $\Delta \propto \rho_{\mathrm{VDS}}^{1/4}\cdot(1+[\mathrm{SSq}]\cdot n/26)$ where the VDS sub-ratio 0.108 places confinement in the sub-threshold regime.

QGP transition (PAPER_1004/1007): At $T > T_c \approx 170$ MeV, the phonon coupling weakens ($\alpha_s \to 0$) and the gap closes, reproducing the deconfinement phase transition observed at ALICE/LHC.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\mathrm{UQFF}} = \mathcal{L}_{\mathrm{GR}} + \mathcal{L}_{\mathrm{SCm}} + \mathcal{L}_{\mathrm{phonon}} + \mathcal{L}_{\mathrm{interaction}}$$

$$\mathcal{L}_{\mathrm{SCm}} = \tfrac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\mathrm{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}$ J/m^3 (matching ρ_{SCm}) and phonon mass $m_{\mathrm{phonon}} = \sqrt{8\lambda}\, v_{\mathrm{SCm}}$.

Nine-sector closure (Session 202):

$\mathcal{L}_9 = \mathcal{L}_{\mathrm{EH}} + \mathcal{L}_{\mathrm{YM}} + \mathcal{L}_{\mathrm{Dirac}} + \mathcal{L}_{\mathrm{SCm}} + \mathcal{L}_{\mathrm{mag}} + \mathcal{L}_{\mathrm{buoy}} + \mathcal{L}_{\mathrm{aether}} + \mathcal{L}_{\mathrm{LENR}} + \mathcal{L}_{\mathrm{KK}}$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\mathrm{gap}} = 5970$ GeV (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\mathrm{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> *Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\mathrm{SSq}]| < 1$ (satisfied by $[\mathrm{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\mathrm{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!} \phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 Axioms $\xrightarrow{\text{DPM + ACP}} \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\mathrm{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\mathrm{NS}} = 0$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.074 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 59, \quad n_{\mathrm{channel}} = 11/26$$

Since $p_{\mathrm{DVP}} = 59$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.074	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\ldots,113\}$	$p_{\mathrm{DVP}} = 59$	PASS Resonant
BSH layers	26 harmonic terms	j = 1…26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element → g_tt = -(1-2μ_s∇(M_s/r)·r/c2) ≡ GR in ε_BSFG\$→\$0 limit	Schwarzschild metric (GR exact)	PDG 2024 / MTW	PASS BSFG reduces to GR
Shapiro time delay	BSFG geodesic → Δt_BSFG ≈ Δt_GR × (1 + ε_correction)	Cassini: Δt/Δt_GR = 1 ± 2.3e-5	Cassini/GR 2003	PASS Within Shapiro bound
Gravitational wave speed v_GW	BSFG: v_GW = c × (1 + \$k_{ }\$2) ≈ c + 10-226 m/s	GW150914 / GW170817:	v_GW/c - 1	< 10-15
Perihelion precession (Mercury)	BSFG adds buoyancy correction $\delta\phi = \kappa$ × ϕ_GR ~ 10-6 arcsec/century	GR prediction: 43.03”/century; observed: 43.1”	GR + obs.	UQFF correction undetectable at current precision

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction Δg ~ 10-6 arcsec/ century to Mercury's perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

§6 Physical Consequences

1. **Dimensional transmutation:** The r^{-3} falloff of T_{s00} (and hence ε) emerges because the 3 compactified dimensions of "effective volume" in the stress-energy tensor are exactly the first three spatial dimensions — giving $V \propto r^3$ and $T_{s00} \propto r^{-3}$.
2. **Kaluza-Klein spectrum:** Each compactified dimension i carries a Kaluza-Klein tower with mass gap $m_i = 1/L_i(r)$. At $r = R_\odot$, $m_i \to \infty$ for all $i \geq 5$ — confirming KK modes are inaccessible at stellar energies.
3. **26D topology:** The BSFG manifold $\mathcal{M}^{26} \cong \mathbb{R}^4 \times T^{22}$ (at macroscopic scales, the extra dimensions form a 22-torus with Planck-scale radii). Its

Euler characteristic $\chi = 0$ (torus topology), consistent with the UQFF non-singular solutions.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1070	Yang-Mills Mass Gap VDS Bridge

7 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_{s26_coupling}.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_{scm_cross_section}.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
kozima_{wstp_kernel}.py	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n * sigma_n^SCm(omega) * Phi_phonon * (F_{U,Bi}/F_U - 1) where sigma_n^SCm(omega,n) = sigma_0 * exp[-(omega-omega_SCm)$^{2/(2*\mathrm{Gamma}}$2)] * (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_{polylog_s26}.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_{wstp\_kernel}.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2$^{\{1\text{-}26\}/(1\text{-}2}${1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_{theta_q26}.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - f_26(q) = Sum_{n=0}^{25} q$^{\{n}$2} / (-q;q)*n^2 - phi_26(q) = Sum*{n=0}^{25} q$^{\{n}$2} / (-q$^{2;q}$2)*n - psi_26(q) = Sum*{n=1}^{26} q$^{\{n}$2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_{pi_uqff}.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_{theta_pi_wstp_kernel}.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n * (1103+26390n) * W_26(n) / C_26 where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

References

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